

Frozen propagation classifier & timestamp-age gate (excerpt, verbatim from frozen source)

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# Timestamp-age gate (C4 rescue, simulator.py):
gate_violation = bool(rescue_c1 and age > age_gate_s + 1e-12)
if gate_violation and not monitor.triggered:
    monitor.triggered = True          # conservative stop
    monitor.trigger_time = t

def decide_label(events, local_pass, baseline_trigger_t, cfg):
    """Pure frozen decision logic. events: dict from extract_events."""
    eps = 1e-9
    trigger_t = events["trigger_t"]
    breach_start_t = events["breach_start_t"]
    margin_breach = breach_start_t is not None
    delayed = (baseline_trigger_t is not None and trigger_t is not None
               and trigger_t > baseline_trigger_t + eps)
    if delayed:
        lo, hi = baseline_trigger_t + eps, trigger_t + eps
        positive_cmd = any(start <= hi and end >= lo
                           for start, end in events["pos_cmd_runs"])
    else:
        positive_cmd = False
    ordered = bool(
        events["t_mismatch"] is not None and delayed and positive_cmd
        and margin_breach and trigger_t <= breach_start_t + eps
        and events["t_mismatch"] <= baseline_trigger_t + eps)

    if events["nan_or_unstable"]:
        label = "UNRESOLVED"
    elif not local_pass:
        label = "ORDINARY_FAILURE"
    elif margin_breach and ordered:
        label = "PROPAGATED"
    elif margin_breach:
        label = "BREACH_UNORDERED"
    else:
        label = "CONTAINED"
```